



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Constituent of Symbiosis International (Deemed University), Pune

Name of Student: Sumit Kumar Mahato

PRN: 25070122195

Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 7

TITLE: :Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

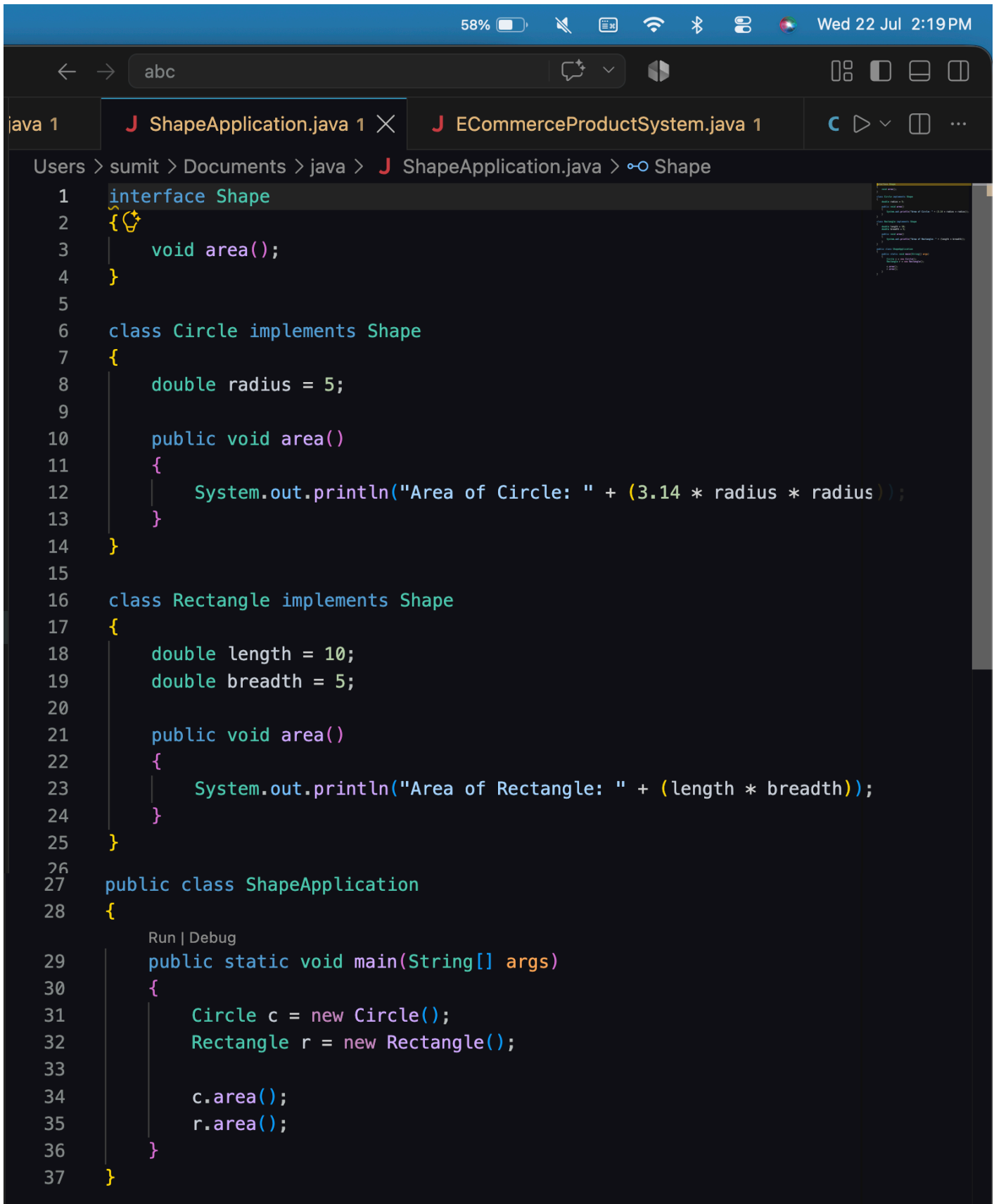
To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Exercise

1. Create a **Shape application** where Circle and Rectangle inherit from Shape and calculate area.



The screenshot shows an IDE window with the following elements:

- Top Bar:** Displays system status (58% battery, network icons) and the date/time (Wed 22 Jul 2:19 PM).
- Tab Bar:** Shows two open files: `ShapeApplication.java 1` (active) and `ECommerceProductSystem.java 1`.
- Breadcrumb:** Indicates the current file path: `Users > sumit > Documents > java > ShapeApplication.java > Shape`.
- Code Editor:** Contains the following Java code:

```
1 interface Shape
2 {
3     void area();
4 }
5
6 class Circle implements Shape
7 {
8     double radius = 5;
9
10    public void area()
11    {
12        System.out.println("Area of Circle: " + (3.14 * radius * radius));
13    }
14 }
15
16 class Rectangle implements Shape
17 {
18     double length = 10;
19     double breadth = 5;
20
21    public void area()
22    {
23        System.out.println("Area of Rectangle: " + (length * breadth));
24    }
25 }
26
27 public class ShapeApplication
28 {
29     Run | Debug
30     public static void main(String[] args)
31     {
32         Circle c = new Circle();
33         Rectangle r = new Rectangle();
34
35         c.area();
36         r.area();
37     }
38 }
```
- Right Margin:** A small, partially visible snippet of code from another file.

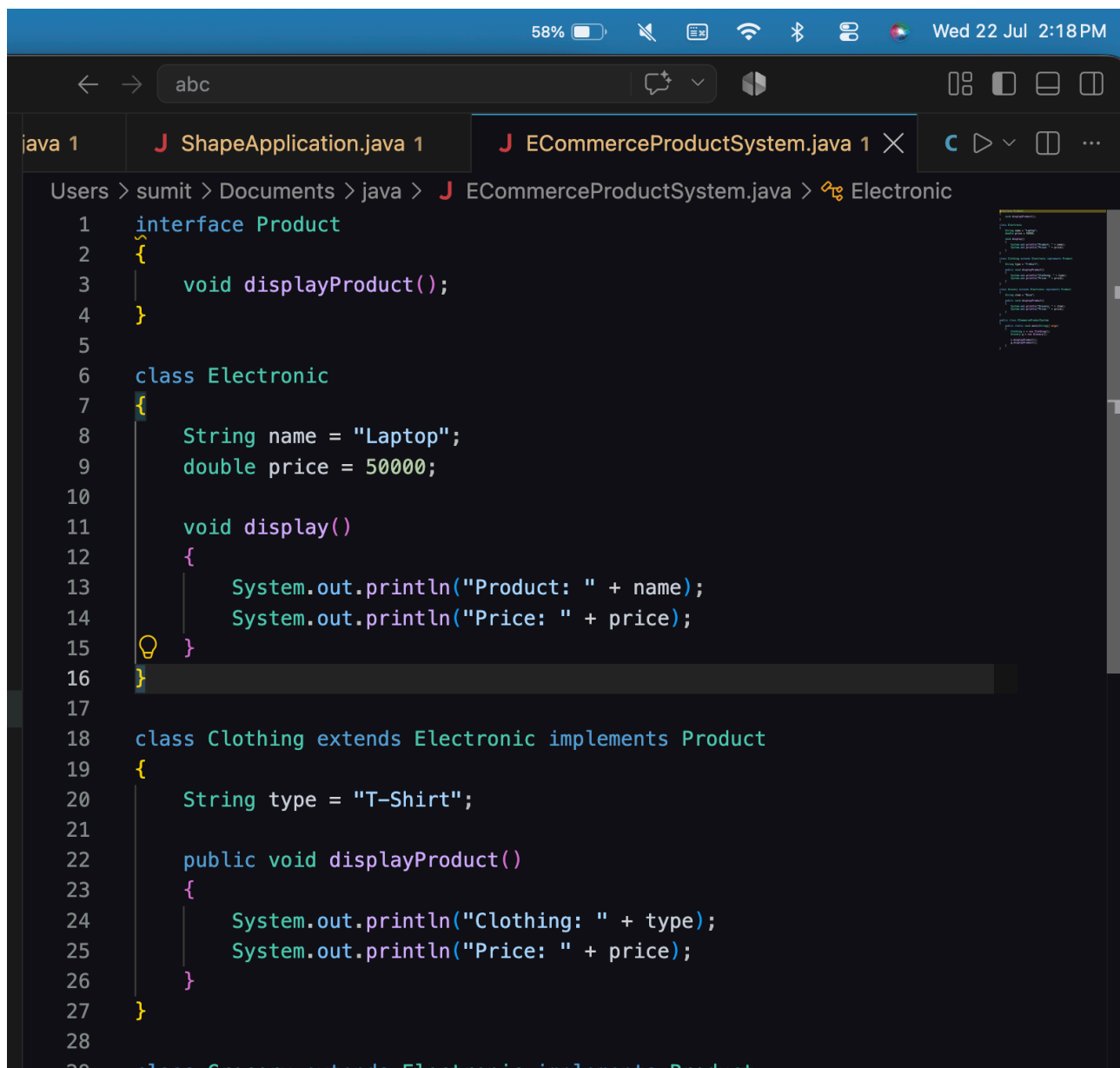
Output:

```
[Running] cd "/Users/sumit/Documents/java/" && javac ShapeApplication.java && java ShapeApplication
Area of Circle: 78.5
Area of Rectangle: 50.0

[Done] exited with code=0 in 1.141 seconds
```

2. Develop a **E-Commerce Product System** where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

Source Code:



```
Users > sumit > Documents > java > ECommerceProductSystem.java > Electronic
1  interface Product
2  {
3      void displayProduct();
4  }
5
6  class Electronic
7  {
8      String name = "Laptop";
9      double price = 50000;
10
11     void display()
12     {
13         System.out.println("Product: " + name);
14         System.out.println("Price: " + price);
15     }
16 }
17
18 class Clothing extends Electronic implements Product
19 {
20     String type = "T-Shirt";
21
22     public void displayProduct()
23     {
24         System.out.println("Clothing: " + type);
25         System.out.println("Price: " + price);
26     }
27 }
28
29 class Grocery extends Electronic implements Product
```

```

29  class Grocery extends Electronic implements Product
30  {
31      String item = "Rice";
32
33      public void displayProduct()
34      {
35          System.out.println("Grocery: " + item);
36          System.out.println("Price: " + price);
37      }
38  }
39
40  public class ECommerceProductSystem
41  {
42      Run | Debug
43      public static void main(String[] args)
44      {
45          Clothing c = new Clothing();
46          Grocery g = new Grocery();
47
48          c.displayProduct();
49          g.displayProduct();
50      }

```

Output:

```

[Running] cd "/Users/sumit/Documents/java/" && javac ECommerceProductSystem.java && java ECommerceProductSystem
Clothing: T-Shirt
Price: 50000.0
Grocery: Rice
Price: 50000.0

[Done] exited with code=0 in 0.306 seconds

```

Conclusion:

The Java programs successfully demonstrated the concepts of inheritance and interfaces using real-world examples. The Shape application showed how Circle and Rectangle can use common properties and calculate their respective areas. The E-Commerce Product System demonstrated the use of interfaces and inheritance for different types of products. These concepts help in achieving code reusability, abstraction, and modularity. Thus, inheritance and interfaces are important features of Java's object-oriented programming.